

$$s = \int_0^t \sqrt{5} \, du = \sqrt{5}t,$$

where we have used u as a dummy variable of integration. Using $t = s/\sqrt{5}$, we obtain a vector equation of the helix as a function of arc length:

$$\mathbf{r}(s) = 2 \cos \frac{s}{\sqrt{5}} \mathbf{i} + 2 \sin \frac{s}{\sqrt{5}} \mathbf{j} + \frac{s}{\sqrt{5}} \mathbf{k}. \quad (4)$$

Parametric equations of the helix are then

$$f(s) = 2 \cos \frac{s}{\sqrt{5}}, \quad g(s) = 2 \sin \frac{s}{\sqrt{5}}, \quad h(s) = \frac{s}{\sqrt{5}}. \quad \equiv$$

The derivative of a vector function $\mathbf{r}(t)$ with respect to the parameter t is a tangent vector to the curve traced by $\mathbf{r}$. However, if the curve is parameterized in terms of arc length s , then $\mathbf{r}'(s)$ is a *unit tangent vector*. To see this, let a curve be described by $\mathbf{r}(s)$, where s is arc length. From (3), the length of the curve from $\mathbf{r}(0)$ to $\mathbf{r}(s)$ is $s = \int_0^s \|\mathbf{r}'(u)\| \, du$. Differentiation of this last equation with respect to s then yields $\|\mathbf{r}'(s)\| = 1$.

9.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-22.

In Problems 1–10, graph the curve traced by the given vector function.

1. $\mathbf{r}(t) = 2 \sin t \mathbf{i} + 4 \cos t \mathbf{j} + t \mathbf{k}; t \geq 0$
2. $\mathbf{r}(t) = \cos t \mathbf{i} + t \mathbf{j} + \sin t \mathbf{k}; t \geq 0$
3. $\mathbf{r}(t) = t \mathbf{i} + 2t \mathbf{j} + \cos t \mathbf{k}; t \geq 0$
4. $\mathbf{r}(t) = 4 \mathbf{i} + 2 \cos t \mathbf{j} + 3 \sin t \mathbf{k}$
5. $\mathbf{r}(t) = \langle e^t, e^{2t} \rangle$
6. $\mathbf{r}(t) = \cosh t \mathbf{i} + 3 \sinh t \mathbf{j}$
7. $\mathbf{r}(t) = \langle \sqrt{2} \sin t, \sqrt{2} \sin t, 2 \cos t \rangle; 0 \leq t \leq \pi/2$
8. $\mathbf{r}(t) = t \mathbf{i} + t^3 \mathbf{j} + t \mathbf{k}$

9. $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + e^t \mathbf{k}$
10. $\mathbf{r}(t) = \langle t \cos t, t \sin t, t^2 \rangle$

In Problems 11–14, find the vector function that describes the curve C of intersection between the given surfaces. Sketch the curve C . Use the indicated parameter.

11. $z = x^2 + y^2, y = x; x = t$
12. $x^2 + y^2 - z^2 = 1, y = 2x; x = t$
13. $x^2 + y^2 = 9, z = 9 - x^2; x = 3 \cos t$
14. $z = x^2 + y^2, z = 1; x = \sin t$
15. Given that $\mathbf{r}(t) = \frac{\sin 2t}{t} \mathbf{i} + (t - 2)^5 \mathbf{j} + t \ln t \mathbf{k}$, find $\lim_{t \rightarrow 0^+} \mathbf{r}(t)$.
16. Given that $\lim_{t \rightarrow a} \mathbf{r}_1(t) = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\lim_{t \rightarrow a} \mathbf{r}_2(t) = 2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$, find:
- (a) $\lim_{t \rightarrow a} [-4\mathbf{r}_1(t) + 3\mathbf{r}_2(t)]$
- (b) $\lim_{t \rightarrow a} \mathbf{r}_1(t) \cdot \mathbf{r}_2(t)$.

In Problems 17–20, find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$ for the given vector function.

17. $\mathbf{r}(t) = \ln t \mathbf{i} + \mathbf{j}, t > 0$
18. $\mathbf{r}(t) = \langle t \cos t - \sin t, t + \cos t \rangle$
19. $\mathbf{r}(t) = \langle te^{2t}, t^3, 4t^2 - t \rangle$
20. $\mathbf{r}(t) = t^2 \mathbf{i} + t^3 \mathbf{j} + \tan^{-1} t \mathbf{k}$

In Problems 21–24, graph the curve C that is described by $\mathbf{r}$ and graph $\mathbf{r}'$ at the indicated value of t .

21. $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 6 \sin t \mathbf{j}; t = \pi/6$
22. $\mathbf{r}(t) = t^3 \mathbf{i} + t^2 \mathbf{j}; t = -1$
23. $\mathbf{r}(t) = 2\mathbf{i} + t\mathbf{j} + \frac{4}{1 + t^2} \mathbf{k}; t = 1$
24. $\mathbf{r}(t) = 3 \cos t \mathbf{i} + 3 \sin t \mathbf{j} + 2t \mathbf{k}; t = \pi/4$

In Problems 25 and 26, find parametric equations of the tangent line to the given curve at the indicated value of t .

25. $x = t, y = \frac{1}{2} t^2, z = \frac{1}{3} t^3; t = 2$

26. $x = t^3 - t, y = \frac{6t}{t + 1}, z = (2t + 1)^2; t = 1$

In Problems 27–32, find the indicated derivative. Assume that all vector functions are differentiable.

27. $\frac{d}{dt} [\mathbf{r}(t) \times \mathbf{r}'(t)]$

28. $\frac{d}{dt} [\mathbf{r}(t) \cdot (t\mathbf{r}(t))]$

29. $\frac{d}{dt} [\mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))]$

30. $\frac{d}{dt} [\mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))]$

31. $\frac{d}{dt} \left[\mathbf{r}_1(2t) + \mathbf{r}_2\left(\frac{1}{t}\right) \right]$

32. $\frac{d}{dt} [t^3 \mathbf{r}(t^2)]$

In Problems 33–36, evaluate the given integral.

33. $\int_{-1}^2 (t\mathbf{i} + 3t^2\mathbf{j} + 4t^3\mathbf{k}) dt$

34. $\int_0^4 (\sqrt{2t + 1}\mathbf{i} - \sqrt{t}\mathbf{j} + \sin \pi t\mathbf{k}) dt$

35. $\int (te^t\mathbf{i} - e^{-2t}\mathbf{j} + te^{t^2}\mathbf{k}) dt$

36. $\int \frac{1}{1 + t^2} (\mathbf{i} + t\mathbf{j} + t^2\mathbf{k}) dt$

In Problems 37–40, find a vector function $\mathbf{r}$ that satisfies the indicated conditions.

37. $\mathbf{r}'(t) = 6t\mathbf{i} + 6t\mathbf{j} + 3t^2\mathbf{k}; \mathbf{r}(0) = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$

38. $\mathbf{r}'(t) = t \sin t^2\mathbf{i} - \cos 2t\mathbf{j}; \mathbf{r}(0) = \frac{3}{2}\mathbf{i}$

39. $\mathbf{r}''(t) = 12t\mathbf{i} - 3t^{-1/2}\mathbf{j} + 2\mathbf{k}$; $\mathbf{r}'(1) = \mathbf{j}$, $\mathbf{r}(1) = 2\mathbf{i} - \mathbf{k}$
40. $\mathbf{r}''(t) = \sec^2 t\mathbf{i} + \cos t\mathbf{j} - \sin t\mathbf{k}$;
 $\mathbf{r}'(0) = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{r}(0) = -\mathbf{j} + 5\mathbf{k}$

In Problems 41–44, find the length of the curve traced by the given vector function on the indicated interval.

41. $\mathbf{r}(t) = a \cos t\mathbf{i} + a \sin t\mathbf{j} + ct\mathbf{k}$; $0 \leq t \leq 2\pi$
42. $\mathbf{r}(t) = t\mathbf{i} + t \cos t\mathbf{j} + t \sin t\mathbf{k}$; $0 \leq t \leq \pi$
43. $\mathbf{r}(t) = e^t \cos 2t\mathbf{i} + e^t \sin 2t\mathbf{j} + e^t\mathbf{k}$; $0 \leq t \leq 3\pi$
44. $\mathbf{r}(t) = 3t\mathbf{i} + \sqrt{3}t^2\mathbf{j} + \frac{2}{3}t^3\mathbf{k}$; $0 \leq t \leq 1$
45. Express the vector equation of a circle $\mathbf{r}(t) = a \cos t\mathbf{i} + a \sin t\mathbf{j}$ as a function of arc length s . Verify that $\mathbf{r}'(s)$ is a unit vector.
46. If $\mathbf{r}(s)$ is the vector function given in (4), verify that $\mathbf{r}'(s)$ is a unit vector.
47. Suppose $\mathbf{r}$ is a differentiable vector function for which $\|\mathbf{r}(t)\| = c$ for all t . Show that the tangent vector $\mathbf{r}'(t)$ is perpendicular to the position vector $\mathbf{r}(t)$ for all t .
48. In Problem 47, describe geometrically the kind of curve C for which $\|\mathbf{r}(t)\| = c$.

Miscellaneous Problems

49. Prove Theorem 9.1.4(ii).
50. Prove Theorem 9.1.4(iii).
51. Prove Theorem 9.1.4(iv).
52. If $\mathbf{v}$ is a constant vector and $\mathbf{r}$ is integrable on $[a, b]$, prove that $\int_a^b \mathbf{v} \cdot \mathbf{r}(t) dt = \mathbf{v} \cdot \int_a^b \mathbf{r}(t) dt$.